

MongoDB Operators – 15 Scenario-Based Practice Questions

Assume the following collections are available: `students` , `employees` , `products` , `orders` , `courses` , and `patients` .

1. Student Scholarship Eligibility

A university wants to identify students eligible for a merit scholarship.

The `students` collection contains:

```
{
  studentId: 101,
  name: "Rahul",
  branch: "CSE",
  cgpa: 8.7,
  attendance: 91,
  year: 2
}
```

Task: Find students whose **CGPA is greater than 8.0** and **attendance is at least 85%** .

Operators: `$gt` , `$gte` , `$and`

2. High-Paid Employees

The HR department wants to identify employees who are either highly experienced or highly paid.

Task: Find employees who have **more than 8 years of experience OR salary greater than ₹80,000** .

Operators: `$gt` , `$or`

3. Electronics Product Filter

An e-commerce company wants to find affordable electronic products.

Task: Find products where:

- Category is "Electronics"
- Price is between ₹10,000 and ₹50,000

Operators: \$eq , \$gte , \$lte , \$and

4. Students from Selected Branches

The placement cell wants a list of students belonging to either **CSE, AI, or Data Science** .

Task: Find students whose branch belongs to these three branches.

Hint: Use the \$in operator.

5. Employees Not in Management

The HR department wants to identify employees who are **not working in the HR or Finance departments** .

Task: Find all employees whose department is neither "HR" nor "Finance" .

Operators: \$nin , \$not

6. Missing Contact Information

The college database administrator wants to find student records where the **email field is missing** .

Task: Find students who do not have an email field.

Operator: \$exists

Then modify the requirement:

Find students whose `phone` field exists.

7. Product Reviews Analysis

An e-commerce website wants to identify products that have received customer reviews.

Suppose a product document contains:

```
{
  name: "Laptop",
  reviews: [
    {user: "Amit", rating: 5},
    {user: "Neha", rating: 4}
  ]
}
```

Task: Find products where the `reviews` field exists and contains at least one review.

Operators: `$exists` , array-related querying

8. Patients Requiring Attention

A hospital wants to identify patients who satisfy **any one** of the following conditions:

- Age is above 65
- Disease is "Diabetes"
- Blood pressure is above 140

Task: Write a query to retrieve such patients.

Operators: `$or` , `$gt` , `$eq`

9. Course Fee Filter

An online learning platform wants to recommend courses that are reasonably priced.

Task: Find courses where the fee is:

- Greater than ₹2,000
- Less than or equal to ₹8,000
- Category is either "AI" or "Data Science"

Operators: \$gt , \$lte , \$in , \$and

10. Salary Range Using \$not

An organization wants to find employees whose salary is **not greater than ₹1,00,000** .

Task: Write a query using the \$not operator rather than directly using \$lte .

Operator: \$not

11. Student Name Pattern

The admission department wants to find students whose names **start with the letter "A"** .

Task: Write a MongoDB query to find these students.

Operator: \$regex

Then modify it to find students whose names **end with "sh"** .

12. Searching Project Titles

A project coordinator wants to find all projects whose title contains the word "AI" anywhere in the title.

For example:

AI-Based Attendance System
Smart Healthcare using AI
AI Chatbot for Students

Task: Write a query using a regular expression to find these projects.

Operator: \$regex

13. Employee Performance Evaluation

The HR department stores employee performance scores:

```
{  
  name: "Ravi",  
  department: "IT",  
  performanceScore: 87  
}
```

Employees are considered high performers if their score is **greater than or equal to 85** .

Task: Find all high-performing employees.

Then find employees whose score is **between 60 and 84** .

Operators: \$gte , \$lt , \$and

14. Inventory Alert Using Multiple Conditions

An e-commerce company wants to identify products requiring immediate restocking.

A product requires restocking when:

- quantity is less than 10 **AND**
- isActive is true

Task: Write a query to find all such products.

Then modify the query so that products are selected when:

quantity is less than 10 **OR** the product is marked as inactive.

Operators: \$lt , \$eq , \$and , \$or

15. Advanced Student Search

The placement cell wants to shortlist students using the following criteria:

A student should satisfy:

- CGPA $\geq$ 8.0
- Attendance $\geq$ 80%
- Branch is either CSE , AI , or Data Science
- The student must have an email address
- The student's name should contain "an" anywhere, case-insensitive

Task: Write a single MongoDB query to retrieve the shortlisted students.

Operators involved:

\$gte

\$in

\$exists

\$regex

\$options

\$and